National Neighborhood Data Archive (NaNDA): Socioeconomic Status and Demographic Characteristics of Census Tract and ZIP Code Tabulation Areas, United States, 2018-2022

NaNDA

Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of socioeconomic and demographic characteristics (SES) by US census tract and zip code tabulation area (ZCTA) for the years 2018-2022. Example measures include population density; population distribution by race, ethnicity, age, and income, income inequality by race and ethnicity; and proportion of population living below the poverty level, receiving public assistance, and female-headed or single parent families with kids. The dataset also contains a set of theoretically derived measures capturing neighborhood socioeconomic disadvantage and affluence, as well as a neighborhood index of Hispanic, foreign born, and limited English.

This user guide covers three separate data files for years 2018-2022 at different geographic scales, which are:

- Census Tract 2010
- Census Tract 2020
- ZIP Code Tabulation Area (ZCTA) 2020

Information about which dataset to use can be found in the Usage Notes section of this document.

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Data Sources

Population values and proportion variables per census tract and ZCTA are from the American Community Survey (ACS) five-year estimates for 2018-2022 (United States Census Bureau, 2023b).

Land area for each census tract or ZCTA comes from the 2010 and 2020 TIGER/Line shapefiles (United States Census Bureau, 2012, 2024b).

Coverage

The dataset contains one observation per census tract or ZCTA in the fifty United States including Alaska, Hawaii, Washington DC and Puerto Rico.

Methodology

To construct this dataset, we extracted key census indicators related to race, ethnicity, age, income, employment, poverty, and home ownership from the ACS 2022 five-year estimate (covering 2018-2022). We merged the variables with each tract's or ZCTA's land area from the 2020 and 2010 TIGER/Line shapefiles for census tracts and ZIP code tabulation areas.

Updates in newer versions of this dataset (2016-2020 and 2018-2022) include:

- A variable for proportion of single-parent households has been added.
- Income quartiles have been updated to be consistent with the income distribution in the United States for 2016-2020. (This was not done in previous versions of this dataset.)
    - Proportion of families with income less than $40,000
    - Proportion of families with income $40,000 to $74,999

  - Proportion of families with income \$75,000 to \$124,999
  - Proportion of families with income \$125,000 or more
- Measures of racial/ethnic income inequality have been created based on a ratio of median household income in White and Black residents; White and Hispanic residents (Zahodne et al., 2024).

The neighborhood socioeconomic indices are based on a principal components analysis, exploratory factor analysis, and confirmatory factor analysis with census tract or ZCTA indicators from the 2018-2022 ACS 5-year estimates. These indices provide a parsimonious set of theoretically-derived factors that capture the shared variance across a broad spectrum of structural socioeconomic characteristics. Results from the factor analysis indicated three separate factors:

- The first factor, which we interpret as Neighborhood Disadvantage, is characterized by high levels of poverty, low family income, and households receiving public assistance income in the neighborhood. Neighborhoods that concentrate the socioeconomically disadvantaged have fewer resources (e.g., healthy food stores, well-maintained parks, good schools, quality medical care) to promote good health (Ross et al., 2001; Ross & Mirowsky, 2001) and are often vulnerable to disinvestment and environmental hazards (Clarke et al., 2014).
- The second factor, which we interpret as Neighborhood Affluence, is characterized by high levels of people with a college education, families with high income, and people employed in professional/managerial occupations in the neighborhood. Affluent neighborhoods are likely to attract a set of institutions (e.g., food stores, places to exercise, well-maintained buildings and parks) that foster a set of norms (e.g., an emphasis on exercise and healthy diets) conducive to good health (Clarke et al., 2014). Distinct from simply being the absence of neighborhood disadvantage, neighborhood affluence is associated with higher levels of social control and leverage over local institutions that can foster social environments that facilitate health (Clarke & Melendez, 2023).
- The third factor represents a higher proportion of Hispanic, foreign born, and people with limited English proficiency in the neighborhood.

Results from the factor analysis were used to inform the creation of the neighborhood socioeconomic mean indices in the dataset:

- The **neighborhood disadvantage** index variable is a mean of the proportion of people with income below federal poverty level, proportion of households receiving public assistance income, and proportion of families with annual income less than \$40,000.
- The **neighborhood affluence** index variable is a mean of the proportion of adults with a college degree, proportion of people employed in professional or managerial occupations, and proportion of families with annual income of \$125,000 or more.

- The **Hispanic Foreign Born Limited English** index variable is a mean of the proportion of people self-identifying as Hispanic, proportion of people who are foreign born, and proportion who don't speak English well.

Further details on the statistical methodology and results of the factor analyses are provided in the technical documentation (Clarke & Melendez, 2023) for these measures.

Usage Notes

Data users should be aware of the differences in the Neighborhood Disadvantage and Affluence measures in the 2016-2020 and 2018-2022 datasets compared to previous years. While they capture similar constructs of concentrated disadvantage or concentrated affluence and may be conceptually comparable, the items included in these measures have changed as the socioeconomic structure of US neighborhoods has evolved over time.

Users wanting to use this dataset alongside measures from NaNDA's 2000-2017 SES datasets should be aware of key differences between how measures are calculated across datasets. For the year 2000, the indicators necessary to build these neighborhood disadvantage and affluence measures came from the decennial census. These variables were moved to the American Community Survey (ACS) in 2005 and removed from the decennial census in 2010 with the introduction of the American Community Survey. In more recent years, they can be found only in ACS five-year estimates.

When constructing measures for 2000-2010, we used data from the 2000 decennial census to represent 2000, data from the 2008-2012 American Community Survey five-year estimate to represent 2010 (the midpoint of the range). Because no data was available for the intervening years, we imputed values for 2001-2009 using a linear interpolation between the two endpoints. For years 2008-2012, 2013-2017, and 2016-2020 the values represent all five years of the ACS estimate range, because they represent the Census Bureau's best estimate for the entire period. The Census Bureau recommends against combining data from overlapping five-year estimates (United States Census Bureau, 2022).

Additionally, new versions of this dataset (2016-2020 and 2018-2022) use United States Census Bureau geography files from 2020 (United States Census Bureau, 2021), while previous versions use United States Census Bureau geography files from 2010 (United States Census Bureau, 2012). The 2018-2022 version does include a version that uses Census Tract 2010 geography.

Census tracts and ZCTAs have changed from 2010 to 2020. Detailed descriptions of the changes to census geography, as well as documentation that may help with longitudinal analysis can be found on the United States Census Bureau's Relationship Files website (United States Census Bureau, 2024a).

Choice of dataset (Census Tract 2010, Census Tract 2020, or ZCTA 2020) depends on the geographic scale of interest and on the Census geography used in any linked dataset. For example, users should use the 2020 Census Tract data if they are linking NaNDA data with survey data that has geographic identifiers at the Census Tract based on 2020 geography. For survey data with geographic identifiers based on 2010 geography, users should use the 2010 Census tract dataset.

Zip Codes and ZIP Code Tabulation Areas

Users should be aware that ZCTAs are not equivalent to ZIP codes. ZIP codes are linear mail delivery routes created by the US Postal Service. ZIP code tabulation areas are spatial features consisting of census blocks grouped by the predominant ZIP code found on the block. In some cases, a location's address is not the same as its ZCTA. For example, some ZIP codes represent single-point addresses such as large post offices or office buildings. Also, the ZIP code for an address may not match its ZCTA if the ZIP code is not the most common ZIP code on the block. See the Census Bureau's ZCTA overview (United States Census Bureau, 2023a) for more information on how ZCTA boundaries are calculated.

Users interested in combining this dataset with ZIP code geocoded data are encouraged to use a ZIP code to ZCTA crosswalk. Such a crosswalk is available at the Heath Resources & Services Administration Data Warehouse (U.S. Department of Health & Human Services, n.d.). An archived copy of the ZIP code to ZCTA crosswalk as well as sample code for merging crosswalk with NaNDA datasets is available in the ICPSR Linkage Library (Chenoweth & Khan, 2024; John Snow, Inc, 2023).

2018-2022 Census Tract Variables (2010 Geography)

Variable	**Label**
tract_fips10	Census Tract FIPS code, 2010
aland10	Census land area, square miles
totpop	Total population
popden	Persons per square mile
phispanic	Proportion of people of Hispanic origin
pnhwhite	Proportion of people non-Hispanic White
pnhblack	Proportion of people non-Hispanic Black
pfborn	Proportion of people who are foreign born
plimeng	Proportion of persons (age 5+) who don't speak English well
ped1	Proportion of persons with Less than a High School Diploma
ped2	Proportion of persons with High School Diploma and/or Some College
ped3	Proportion of persons with Bachelors Degree or Higher
pfamínclt40k	Proportion of families with income less than $40,000
pfamincge40lt75k	Proportion of families with income $40,000 to $74,999
pfamincge75lt125k	Proportion of families with income $75,000 to $124,999
pfamincge125k	Proportion of families with income $125,000 or more
pnvmar	Proportion of People 15+ Never Married
p18yr_	Proportion of population under 18 yrs
p18_29	Proportion of population 18-29
p30_39	Proportion of population 30-39 yrs
p40_49	Proportion of population 40-49 yrs
p50_69	Proportion of population 50-69 yrs
pge70	Proportion of population 70+ yrs
punemp	Proportion of age 16+ civilian labor force unemployed
pprof	Proportion of civilians employed in Management, business, science, and arts occupations
ppov	Proportion people w/ income in the past 12 months below poverty level
ppubas	Proportion of households with public assistance income or food stamps
pfhfam	Proportion female-headed families w/ kids
psngpnt	Proportion single parent families w/ kids
pownoc	Proportion Owner Occupied Housing Units
affluence	mean of ped3_ pfamincge125k pprof
disadvantage	mean of pfaminclt40k ppov ppubas
ethnicimmigrant	mean of phispanic pfborn plimeng
medfaminc	Median Family Income
medfaminc_nhwhite	Median Family Income (White, Not Hispanic Or Latino Householder)
medfaminc_black	Median Family Income (Black Or African American Alone Householder)
medfaminc_hispanic	Median Family Income (Hispanic Or Latino Householder)
ratio_medfaminc_nhwtob	Ratio Median Family Income, NH-White / Black
ratio_medfaminc_nhwtohisp	Ratio Median Family Income, NH-White / Hispanic

2018-2022 Census Tract Variables (2020 Geography)

Variable	**Label**
tract_fips20	Census Tract FIPS code, 2020
aland	Census land area, square miles
totpop	Total population
popden	Persons per square mile
phispanic	Proportion of people of Hispanic origin
pnhwhite	Proportion of people non-Hispanic White
pnhblack	Proportion of people non-Hispanic Black
pfborn	Proportion of people who are foreign born
plimeng	Proportion of persons (age 5+) who don't speak English well
ped1	Proportion of persons with Less than a High School Diploma
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ppubas	Proportion of households with public assistance income or food stamps
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pownoc	Proportion Owner Occupied Housing Units
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disadvantage	mean of pfaminclt40k ppov ppubas
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medfaminc	Median Family Income
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medfaminc_black	Median Family Income (Black Or African American Alone Householder)
medfaminc_hispanic	Median Family Income (Hispanic Or Latino Householder)
ratio_medfaminc_nhwtob	Ratio Median Family Income, NH-White / Black
ratio_medfaminc_nhwtohisp	Ratio Median Family Income, NH-White / Hispanic

2018-2022 ZCTA Variables (2020 Geography)

Variable	**Label**
zcta20	zip code tabulation area
aland	Census land area, square miles
totpop	Total population
popden	Persons per square mile
phispanic	Proportion of people of Hispanic origin
pnhwhite	Proportion of people non-Hispanic White
pnhblack	Proportion of people non-Hispanic Black
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ratio_medfaminc_nhwtob	Ratio Median Family Income, NH-White / Black
ratio_medfaminc_nhwtohisp	Ratio Median Family Income, NH-White / Hispanic

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